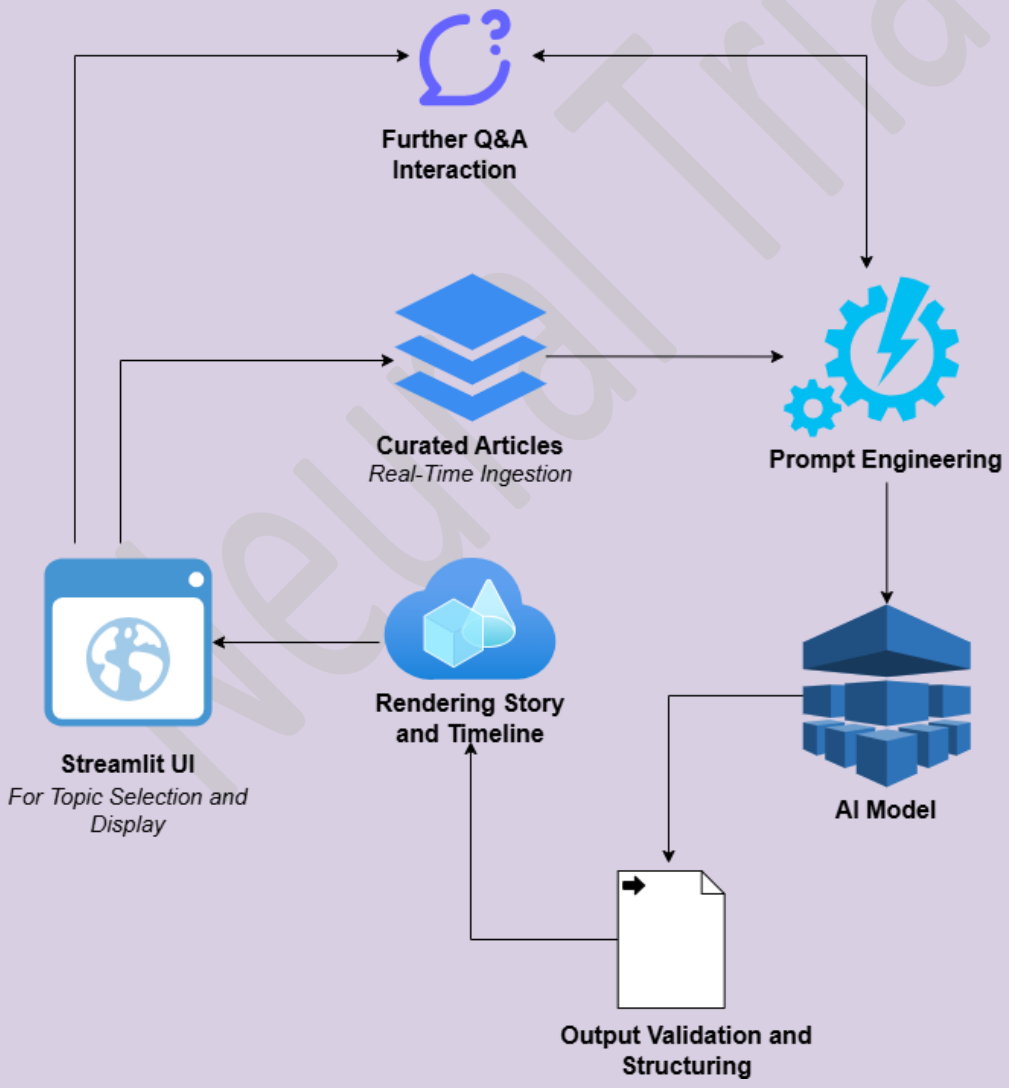
ArcSentry AI - Architecture Document

1. Overview

ArcSentry AI is a Generative AI-based system that converts fragmented news into structured, interactive story intelligence. It enables users to quickly understand complex, evolving topics through a unified narrative, timeline, and contextual insights.

The system is designed to reduce information overload, improve comprehension speed, and enhance user engagement.

2. Architecture Design



Neural Triads

3. Data flow

1. User selects a topic
2. System loads structured article data
3. Prompt is generated
4. Request sent to AI model
5. Structured output returned
6. Output parsed and validated
7. UI renders story and timeline
8. User interacts via Q&A

4. Technical Stack

Frontend: Streamlit
Backend: Python
LLM: Google Gemini API
State Management: Streamlit session_state
Visualization: Custom HTML timeline

5. Scalability

- Supports integration with real-time news sources
- Extendable with vector databases and retrieval pipelines
- Scales across domains and large user bases

6. Key Strengths

- Structured output ensures reliable rendering
- Modular design supports extensibility
- Context-aware AI reduces hallucination
- Visual timeline improves comprehension
- Optimized for performance and usability

7. Conclusion

ArcSentry AI combines Generative AI with structured processing and intuitive visualization to transform news consumption into an insight-driven experience. It enables a shift from fragmented reading to fast, structured understanding.

Neural Triads